



Together Towards a Better Future

معاً نحو غدٍ أفضل

COLLEGE OF HEALTH SCIENCES

Bachelor of Science in Nursing

PML-2183 Clinical Pharmacology for Nursing

COURSE SYLLABUS

Spring Semester 2024-2025

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PML-2183 Clinical Pharmacology for Nursing

1. Course Name: PML-2183 Clinical Pharmacology for Nursing

2. Credit Hours (Lecture-Practical-Credits): Three (3-0-3)

3. Prerequisites and/or Co-Requisites:

Prerequisite: None

Co-requisites: None

Equivalent Course: None

4. Semester: Spring Semester, Academic Year, 2023-2024

5. Name and Contact Information of Faculty:

a. Faculty Name: Dr. Abdelhameed Elshenawy

b. Office: 6

c. Email: abdelhameed@uof.ac.ae

d. Phone: 405

e. Office Hours: Tuesday, 11:00am – 12:00pm

6. Course Description (Catalog Description):

This course introduces the foundational and advanced concepts of clinical pharmacology within the healthcare context, focusing on the interaction between drugs and body systems. Students will explore pharmacodynamics and pharmacokinetics, emphasizing drug interactions and their processes in the body. Through interactive learning activities, Lecture, discussions, Video presentation and Flipped Classroom. By the end of the course, students will demonstrate the ability to apply rational decision-making in clinical pharmacology, develop practical skills in drug administration, and enhance their readiness to address real-world challenges in pharmacological care.

7. Course Learning Outcomes (CLOs):

Upon completion of the course, the student will be able to:

Course Learning Outcomes	*Bloom's Taxonomy	**QFE Alignment	***Linkage and Contribution of CLOs to PLOs	
			Linkage	Extent of Contribution
1. Apply the allied concepts of clinical pharmacology in the health care context.	3	K, AR	1	M
2. Explain the concepts of pharmacodynamics and pharmacokinetics processes associated with drug interaction in the body systems.	4	K, AR	1	M
3. Interpret the classification of drugs and their effect on the body systems.	4	AR	1	M
4. Integrate the methods of drug administration to avoid adverse effect.	6	AR, RC	1, 4	M
5. Determine the legal and professional issues related to drug administration	3	AR, RC	1, 3	M
LEGEND: 1. *Bloom's Taxonomy = Remember (1); Understand (2); Apply (3); Analyze (4); Evaluate (5); Create (6); Affective Domain (AD); Psychomotor Domain (PD). 2. **QFE Alignment = Knowledge (K); Skills (S); <u>Aspects of Competence</u> : Autonomy and Responsibility (AR); Role in Context (RC); Self-Development (SD). 3. ***Linkage and Contributions of CLOs to PLOs = High (H); Medium (M); Low (L). Note: Please refer to item 8: The Program Learning Outcomes of BSN to determine the linkage and extent of contribution of CLOs.				

8. Program Learning Outcomes

Upon completion of the Program, the student will be able to:	
1	Demonstrate a sound knowledge of the essential concepts and theories related to the nursing practice.
2	Perform nursing procedures, and operate technology and health information systems skillfully in the delivery of optimal nursing care to individuals, families and communities.
3	Implement professional behavior consistent with the code of ethics, as well as effective communication strategies to deliver optimal holistic person-centered care to individuals, families and communities.
4	Apply the nursing process, supported by evidence-based practice, to develop a holistic nursing care plan that ensures safe, culture-sensitive nursing care.
5	Develop a research proposal and disseminate research findings to enhance the nursing practice.
6	Differentiate amongst different quality improvement interventions to resolve clinical nursing practice problems through innovation and entrepreneurship.
7	Demonstrate leadership skills during collaboration with health care providers in the delivery of quality health care services to the individuals, families and communities.

9. Professional Certification Qualification (PCQ) Mapping

Applicability of PCQ	Name of PCQ (if Applicable)	Mapping of PCQ to CLO (Refer to the CLOs in item 7)
☒ Applicable		
☒ Not applicable		

10. Schedule of Delivery of Course Topics, Contents, and Assessments

Week	Lecture Topics and Contents	Chapters	Teaching and Learning Methodologies (if required, include laboratory, studio, external visit, and other non-lecture sessions)	Methods of Assessment, Out of Class Assignments, and Due Dates
1-3	General principles of Pharmacology: - Objectives & definitions - Terminology, and general introduction - Pharmacokinetics: Absorption, Distribution, Metabolism & Elimination - Pharmacodynamics - Adverse Drug Reactions or ADRs - Drug Interactions - Drug Calculations	1, 2	- Course Orientation: to include Policy on Academic Integrity, Academic Paper, & APA Referencing Style - Lecture and discussion - Video presentation - Blended Learning – Flipped Classroom - Encourage discourse of the topic - Problem-solving activities - Answering critical thinking questions - Debriefing and recapping	Week 3: Quiz 1
4	Drugs affecting the autonomic nervous system - Introduction - Adrenomimetic drugs - Adrenoreceptor antagonists - Cholinomimetics - - Cholinergic blocking drugs - Cholinesterase inhibitors - Local anesthetics - Muscle relaxants	4	- Lecture and discussion - Video presentation - Blended Learning – Flipped Classroom - Encourage discourse of the topic - Problem-solving activities - Answering critical thinking questions - Debriefing and recapping	
5 & 6	Antimicrobial - Classification of antibiotics - Antiviral Drugs - Fungal infection	5	- Lecture and discussion - Video presentation - Blended Learning – Flipped Classroom - Encourage discourse of the topic - Problem-solving activities	Week 5: Quiz 2

Schedule of Delivery of Course Topics, Contents, and Assessments (*continued*)

Week	Lecture Topics and Contents	Chapters	Teaching and Learning Methodologies (if required, include laboratory, studio, external visit, and other non-lecture sessions)	Methods of Assessment, Out of Class Assignments, and Due Dates
			○ Answering critical thinking questions ● Debriefing and recapping	
7	Ant-inflammatory Drugs ● Inflammatory processes. ● Non-steroidal anti-inflammatory agents. ● Non-steroidal anti-inflammatory drugs ● Naproxen ● Cyclo-oxygenase pathway 2 inhibitors ● Celecoxib ● Aspirin ● Paracetamol ● Antihistamine medication. ● Steroid drugs	6	● Lecture and discussion ● Video presentation ● Blended Learning – Flipped Classroom ○ Encourage discourse of the topic ○ Problem-solving activities ○ Answering critical thinking questions ● Debriefing and recapping	
8 & 9	Drugs affecting Cardiovascular system ● Drugs used in heart failure ● Antihypertensive agents ● Antianginal agents ● Antiarrhythmic agents ● Anticoagulants fibrinolytic drugs, ● Antiplatelet Drugs	7, 10	● Lecture and discussion ● Video presentation ● Blended Learning – Flipped Classroom ○ Encourage discourse of the topic ○ Problem-solving activities ○ Answering critical thinking questions ● Debriefing and recapping	Week 8: Mid-Term Examination
10 & 11	Drugs affecting the Respiratory system Asthma ● Drugs used in treating asthma ● Beta 2 adrenoreceptor agonists ● Methylxanthines ● Ipratropium ● Inhaled corticosteroids ● Drugs used in COPD ● Inhaled long-acting beta 2 agonists (LABAs) ● Long-acting muscarinic antagonists (LAMAs) ● Phosphodiesterase 4 inhibitors	8	● Lecture and discussion ● Video presentation ● Blended Learning – Flipped Classroom ○ Encourage discourse of the topic ○ Problem-solving activities ○ Answering critical thinking questions ● Debriefing and recapping	Week 10: Quiz 3
12	Drugs affecting the endocrine system Drugs used in Diabetes Mellitus ● Treatment with insulin ● Oral hypoglycemic agents ● Metformin ● Sulfonylureas ● Meglitinides ● Thiazolidinediones (Glitazones) ● Alpha-glucosidase inhibitors Drugs used in hyper and hypothyroidism Drugs used in hyper and hypoparathyroidism Drug Therapy for Addison's Disease and Cushing's Disease		● Lecture and discussion ● Video presentation ● Blended Learning – Flipped Classroom ○ Encourage discourse of the topic ○ Problem-solving activities ○ Answering critical thinking questions ● Debriefing and recapping	Week 12: Written Assignment
13	Mental health drugs Medicine management of anxiety ● Barbiturates ● Benzodiazepines ● Beta-adrenergic blockers ● Azaspirodecenediones ● Antipsychotics ● Antidepressant agents	11	● Lecture and discussion ● Video presentation ● Blended Learning – Flipped Classroom ○ Encourage discourse of the topic ○ Problem-solving activities	Week 13: Quiz 4

Schedule of Delivery of Course Topics, Contents, and Assessments (*continued*)

Week	Lecture Topics and Contents	Chapters	Teaching and Learning Methodologies (if required, include laboratory, studio, external visit, and other non-lecture sessions)	Methods of Assessment, Out of Class Assignments, and Due Dates
	Medicine management of depression • Tricyclic antidepressants • Monoamine oxidase inhibitors • Selective serotonin re-uptake inhibitors Anticonvulsants drugs Antiparkinsonian drugs		○ Answering critical thinking questions • Debriefing and recapping	
14	Chemotherapy drugs • Alkylating agents • Antimetabolites • Plant alkaloids • Topoisomerase inhibitors	7, 10	• Lecture and discussion • Video presentation • Blended Learning – Flipped Classroom ○ Encourage discourse of the topic ○ Problem-solving activities ○ Answering critical thinking questions • Debriefing and recapping	
15	Legal and professional issues • Patient's rights • Issues related to drug administration • Incidence report	8	• Lecture and discussion • Video presentation • Blended Learning – Flipped Classroom ○ Encourage discourse of the topic ○ Problem-solving activities ○ Answering critical thinking questions • Debriefing and recapping	
16	Final Examination			Week 16: Final Written Examination

11. Assessments Information: Methods and Dates of Examinations and Relative Weights

Assessment Task	Weight (%)	Group/Individual	Due Date of Exams and Assessments
I. Coursework	40		
A. QUIZZES	20		
1. Quiz 1	5	Individual	3 rd Week
2. Quiz 2	5	Individual	5 th Week
3. Quiz 3	5	Individual	10 th Week
4. Quiz 4	5	Individual	13 th Week
B. Assignment	15		
a. A Written assignment	15	Group	12 th Week
b. Brochure	5	Individual	12 th Week
II. Mid-Term:	20		
1. Mid-Term Written Examination	20	Individual	8 th Week
III. Final Examination:	40		
1. Final Written Examination	40	Individual	16 th Week
Total	100		

12. Assessment Instruments and CLO Mapping

CLO	Coursework (40%)						Midterm Exam (20%)	Final Exam (40%)	Total (100%)
	Quiz1 (5%)	Quiz 2 (5%)	Quiz3 (5%)	Quiz 4 (5%)	Written assignment (15%)	Brochure (5%)			
1	3						6	4	13
2	2	2	3		15	5	10	3	40
3		3					4	13	20
4			2	5				14	21
5								6	6
Total	5	5	5	5	15	5	20	40	100

13. Course Textbook/s, Recommended Supplemental Materials and Learning Management System

Description	
• Prescribed Textbook/s	Paul Barber and Deborah Robertson (2020) Essentials of Pharmacology for Nurses. McGraw-Hill Education. ISBN-13: 978-0-33-523404-2
• Recommended Readings (Textbooks/Journals/Articles)	Edmunds, M. W. (2015). Introduction to Clinical Pharmacology-E-Book. Elsevier Health Sciences. Geraldyn Frandsen, Sandra Smith Pennington Abrams' Clinical Drug Therapy: Rationales for Nursing Practice, Tenth Edition
• Learning Management System	Smart Projector and Desktop (installed in each classroom)



Written Assignment and Assessment Rubric

Course Title:

Course Code:

Student Name:

Student ID:

Date:

The project is divided into two parts: a written report and a brochure.

Part 1- A Written Report

Student's Instructions:

- Students are required to **work in groups of a maximum of 5 students/group** to write a pharmaceutical informational report for one of the following drugs classifications (Table1):

Table 1: Suggested drugs classifications

Classification	Name of the Drug	Classification	Name of the Drug
Anti-Inflammatory Drugs	1. Loratadine 2. Ibuprofen 3. Zileuton 4. Hydrocortisone	Drugs for Heart Failure and Dysrhythmias	1. Enalapril 2. Nitroglycerin 3. Digoxin 4. Dopamine 5. Atropine
Diuretic and Hypertension Drugs	1. Furosemide 2. Enalapril 3. Losartan 4. Atenolol 5. Spironolactone	Blood Clotting Drugs	1. Heparin 2. Aspirin 3. Clopidogrel 4. Warfarin 5. Alteplase (TPA)

- Report Components include:

- 2.1 Generic name
- 2.2 The trade names available in UAE
- 2.3 Therapeutic Classification
- 2.4 Pharmacological classifications
- 2.5 Names of other drugs in the same pharmacological classification
- 2.6 Indication/s
- 2.7 Route of administration
- 2.8 Available forms
- 2.9 Required doses for adult
- 2.10 Mechanism of action (Action. How does this medication work in the body?)
- 2.11 Preparation requirements (if applicable)
- 2.12 Adverse reactions. List life threatening ones first. Then list the most common.
- 2.13 Contraindications. List the conditions which would prohibit use of this drug.
- 2.14 Food/Drug interactions and/or incompatibility. List.
- 2.15 Nursing Implications: Assessment: What does the nurse do to safely assess and monitor with this med?
- 2.16 Implementations: What are the nursing implications before administering the drug (focus on patient safety)?
- 2.17 Patient teaching: What does the nurse need to teach the patient?
- 2.18 Labs: If applicable, write in which specific lab needs to be monitored, and why correlate.
- 2.19 Clinical Research study on the effectiveness of the chosen drug ((published not more than 10 years ago)

3. Report Structure includes:
 - 3.1 Cover/title page (Name of Institution, name of campus, title of report, course name and code, your name, date of submission)
 - 3.2 Introduction
 - 3.3 Body
 - 3.3.1 Background/literature review
 - 3.3.2 Analysis of information
 - 3.3.3 Discussion
 - 3.4 Conclusion
 - 3.5 Recommendations (UAE context)
 - 3.6 References (APA style) (Minimum number of academic/scientific references is 5).
4. Report format (APA Style) includes:
 - 4.1 The main text should be in 12-point Times New Roman, double spaced, Indent the first line of every paragraph 0.5 inches.
 - 4.2 Please do not use double-spacing between paragraphs.
 - 4.3 Print on A4 paper leaving a 1-inch margin all around.
 - 4.4 Any figures or tables should have a caption. Figure captions should be below the figures and table titles should be centered and above the tables.
 - 4.5 Page numbers on upper right corner.
 - 4.6 Project Report in right foot note.
 - 4.7 Information in your report should be based on scholarly work. Literature review must be done to identify recent evidence-based information.
 - 4.8 Make sure your work is grammatically and structurally correct.
5. In this project, students are required to submit a project written report following the evaluation criteria below in the Written Assignment Rubric. The written report accounted 15 % of the total course grade.
6. A detailed action that includes the contribution of each group member, should be submitted with the report.
7. Each student should submit a soft copy of the written report **along with an individual signed cover page.**
8. The submission will be by **week 12**
9. If you have any concerns or questions, please raise them early (before 2 weeks of submission) to the Faculty / Lecturer.

Part 2- A Brochure

Student's Instructions:

1. Each student is expected to submit **an individually** prepared brochure following the evaluation criteria below in the Presentation/Brochure Rubric. The is accounted 5 % of the total course grade.
2. Content include Drug name, therapeutic Classification, Indication/s, Route of administration, required doses for adult, brief Mechanism of action, Preparation requirements (if applicable), Adverse reactions (List life threatening ones first, then list the most common), Contraindications, Food/Drug interactions.
3. Submit the References and resources on a separate sheet attached to the poster, and submit the entire package by **12th week.**

Written Assignment and Assessment Rubric (*continued*)

Criteria	Possible Marks				Mark
Introduction	0	1-2	3-4	5	
	Not provided a definition or good background on the condition. Major errors or inconsistencies in content or terminology	Made an attempt to provide a definition and / or identify the DSM classification but there are minor inconsistencies. Confusing to understand	Fairly covers criteria but lacks full detail or flow	All criteria covered well in clear, concise manner and uses appropriate medical terminology	
Main Body	0-10	11-20	21-30	31-40	
Drug classification, indications, side effects, routes of administrations and mechanism of action, lab and research /food and drug interactions.	Not all criteria covered and/or not clearly developed. Major inconsistencies present. Does not demonstrate an understanding of the condition.	Covers some of the criteria but further development / discussion required. Further linkages needed. Does not fully demonstrate an understanding of the condition and impact on treatment modalities	Fairly well written for most of the criteria. Demonstrates a good understanding of the condition, treatment modalities and nursing management. Shows some consideration of the impact the condition has on patient's life	Well structured, logical. Very good to excellent coverage of all criteria demonstrating full understanding of condition, treatment modalities and nursing management	
Summary/Conclusion	0	1-2	3-4	5	
	Not identified or not discussed	Identified but lacks detail	Identified and fairly well discussed	Well covered	
Research	0	1-2	3-4	5	
	No evidence of research conducted	Some attempt made no strong arguments supplied. Only one source used	Covered fairly well but arguments not fully developed. Limited range of sources	Well researched. Good strong arguments made. Links theory to practice. Wide range of resources	
Referencing	0	1-2	3-4	5	
	Sources not cited	Some attempt made but major inaccuracies	Fairly good attempt with minor inaccuracies	Well referenced using APA	
Formatting, Language and Grammar	0	1-2	3-4	5	
	Did not follow formatting instructions. Errors in Grammar and spelling	Poor presentation and / or major spelling / grammar errors	Good presentation but minor spelling / grammar errors	Excellent presentation with no errors	
				Total Marks = 65	/65
Instructor Feedback Instructor Name: _____ Instructor Signature: _____ Date: _____					

Presentation/ Brochure Rubric (*continued*)

Criteria	PERFORMANCE LEVEL				Mark
	0	1	2	3	
TASK MANAGEMENT Organization of approach (Action Plan submitted)	Task conducted in total disorder. No action plan submitted	Task conducted in some disorder as evidenced by performance or action plan	Task conducted in adequate manner. Action plan submitted.	Task conducted in a well-organized manner as evidenced by performance and action plan	
Working with others	No attempt at cooperation	Minimal attempt to work as team member	Works as team member with prompting	Works effectively as team member	
DESIGN Use of illustrations	Illustrations used are unrepresentative of subject	Illustrations used have several minor inaccuracies or a major inaccuracy	Illustrations are used with few minor inaccuracies	Illustrations used accurately	
Use of written information	Information inappropriate & inaccurate	Information has several minor inaccuracies or a majority inaccuracy	Information used with few minor inaccuracies	Information used in appropriate form	
METHOD Use of appropriate medium to present information	Method entirely inappropriate.	Some attempt made to use appropriate medium but major errors apparent	Attempt made to use appropriate medium but minor errors apparent	Appropriate medium used for presentation. No errors apparent.	
CONTENT Selection and retrieval of appropriate materials (References and resources to be submitted on separate sheet attached to the poster)	No references or evidence of research presented	Limited references or range of resources identified but not evaluated	Relevant information from full range of sources presented but lacks evaluation	Used full range of sources to find relevant information and this is appropriately evaluated	
Use of appropriate vocabulary	Vocabulary inappropriate	Some attempt made to use appropriate vocabulary but with major errors apparent	Attempt made to use appropriate vocabulary but with minor errors apparent	Vocabulary fully appropriate	
Depth of content	Content superficial and without relevance to the subject	Some attempt made to attain correct depth but major errors apparent	Attempt made to attain correct depth but with minor errors apparent	Depth appropriate to target audience	
GENERAL Overall lay-out	No thought to lay-out. Information unclear & in concise	Poster eye-catching but information unclear	Poster not eye-catching but information clear & concise	Eye catching lay-out, informative, clear & concise	
	0	4	6	8	
Communication and Body Language	Inappropriate communication and body language	Avoiding eye contact, reading from paper/presentation	Not standing up straight, somewhat accurate in communication	Natural delivery, fluent and appropriate vocabulary	
Total Marks = 35					/35
Instructor Feedback					
Instructor Name: _____ Instructor Signature: _____ Date: _____					